Book strip

Science · LAUNCH · Autumn 2 · Week 1 · SCI-LAUNCH-A2-W01-L3

Master: matching chromosome sets and repair

I am learning to justify the cell-cycle order and explain how matching chromosome sets support growth and repair.

Success: justify the order; explain matching sets; link new cells to growth and repair.

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Route card

Staff: cut these apart before the lesson. Give one route card. Other routes remain available.

Supported



Supported

Supported route card with a model cell

Order interphase, mitosis and cytokinesis. Say why copying must come first. [3]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Independent



Independent

Independent route card with a model cell

Order the six named stages. Mark the four mitosis phases and justify why interphase comes first. [4]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Aspire



Aspire

Aspire route card with a model cell

Predict what happens if copied chromosomes separate unevenly. Explain why a correct final split cannot fix the sets. [3]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Choice



Choice

Choice route card with a model cell

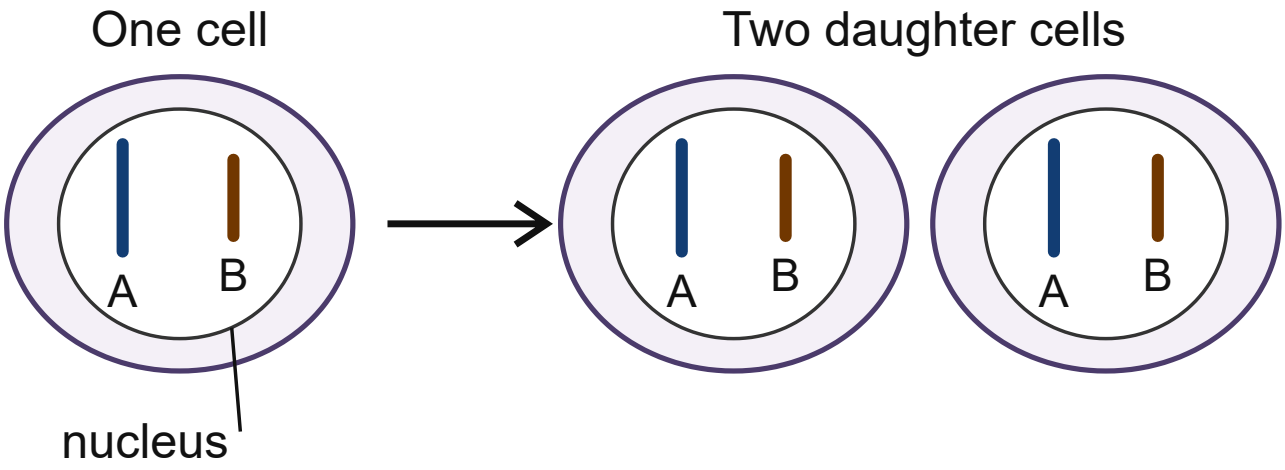
Choose the order of interphase, mitosis and cytokinesis. Give one reason why copying comes first. [3]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Made by Matt

Reference A matching chromosome sets



Compare the full A and B set in both daughter cells. Each new cell has a matching set.

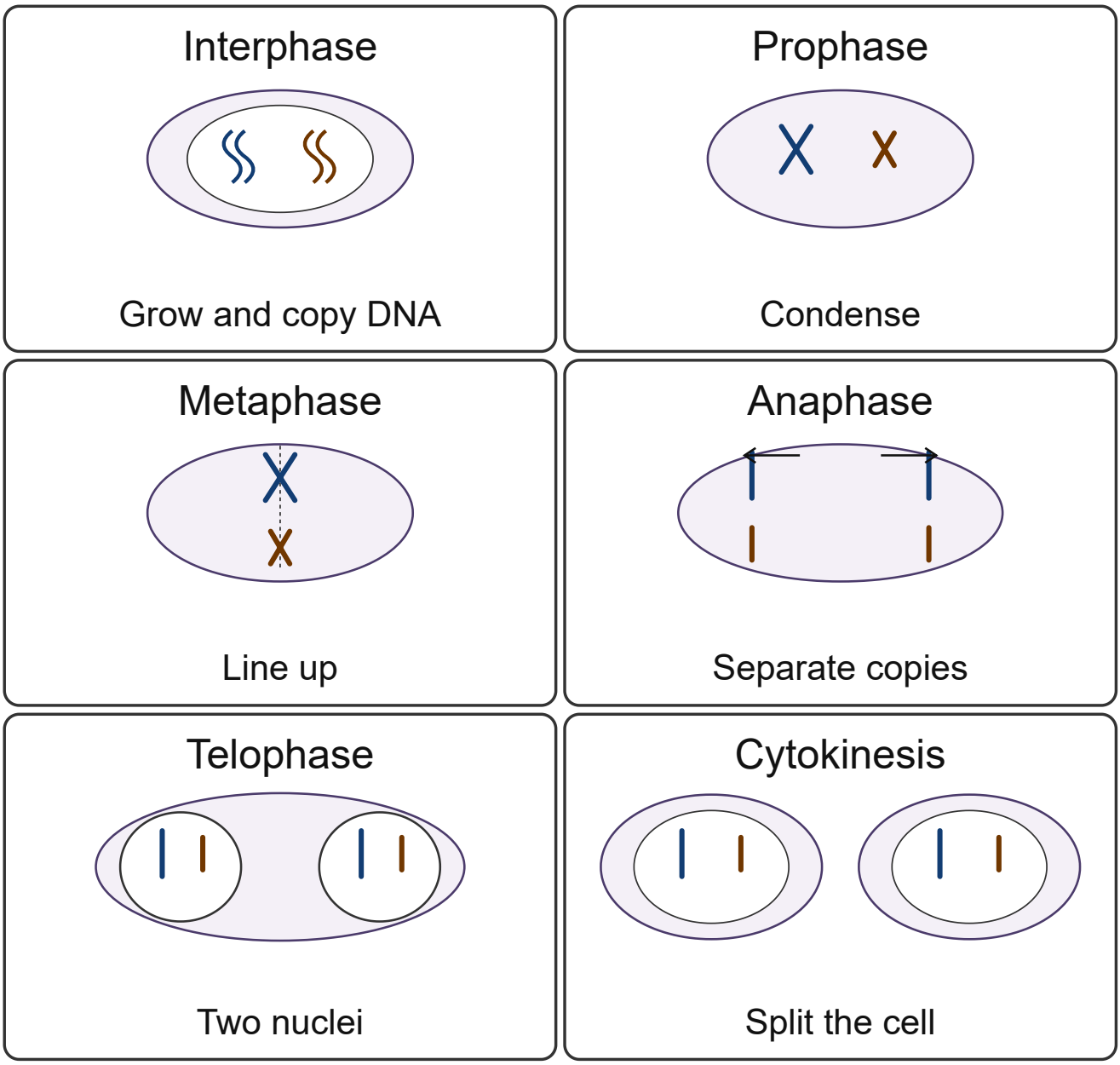
Reference only. Use labels as well as colours. The symbols are simplified chromosome types, not a full picture of real DNA.

Step	What changes
Interphase	The cell grows and copies its DNA.
Mitosis	Copied chromosomes separate into two nuclei.
Cytokinesis	The cell splits into two daughter cells.

In the number case, six chromosomes in the starting body cell means six in each daughter after ordinary mitosis and cell division. A fresh case starting with eight gives eight in each daughter.

Made by Matt

Reference B the six named stages



Interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Four middle phases belong to mitosis.

Reference only. Interphase comes before mitosis. Cytokinesis follows the formation of the two nuclei. The four named mitosis phases are prophase, metaphase, anaphase and telophase.

Made by Matt

Catch-up sheet absence or access only

Use only after absence or an access barrier. This replaces a starting step; it is not added missed work.

Interphase: the cell grows and copies DNA. Mitosis: copied chromosomes separate into two nuclei. Cytokinesis: the cell divides.

Use Reference A to put the three main stages in order. Check both new sets. Then join the current model or You do task 1.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

Made by Matt

Choice task sheet

Select an answer and give a short reason. Use Reference A and B.

1. Choose the order of interphase, mitosis and cytokinesis. Give one reason why copying comes first. [3] Staff I ☐ S ☐ C ☐

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2. Each new body cell needs A and B. Choose one copy of each or both A copies at one end. Explain your choice. [2] Staff I ☐ S ☐ C ☐

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3. Choose new matching cells or uneven chromosome sets to help replace damaged skin cells. Use copy, separate and split to show why. [3] Staff I ☐ S ☐ C ☐
